

CBXFEL DESIGN, PRODUCTION, ASSEMBLY, TESTING AND INTALLATION STATUS

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Abstract

Use of a cavity-based X-ray free electron laser (CBXFEL) is potentially a way to dramatically improve the stability and coherence of existing XFELs. CBXFEL consists of a low-emittance electron source, a magnet system with several undulators and chicanes, and an X-Ray cavity. The X-Ray cavity stores and circulates X-ray pulses for repeated FEL interactions with electron pulses until the FEL reaches saturation. The CBXFEL is expected to operate using 9.831 keV photons from LCLS, using synthetic diamonds as cavity Bragg mirrors. The LCLS copper linac will deliver two electron bunches 624 RF buckets apart, resulting in a total X-ray cavity length of 65500.87 mm. The final X-ray cavity design, assembly, and installation status will be presented in this paper.

INTRODUCTION

The mechanical system for the rectangular x-ray cavity at LCLS-II consists of four diamond crystals. The first two of these crystals (C4 and C1) reflect the X-ray pulse into (C4) and out (C1) of the undulator line when operating the FEL with the x-ray cavity in CBXFEL mode. These optics need to be inserted on the undulator axis and are accompanied by additional x-ray and electron beam diagnostics (X42, X42, X11, X10, BODs) [1-3].

Crystals C2 and C3 reflect the X-ray pulse in and out of the return line.

The mechanical system also includes five diagnostic stations (Stations C, D, E, F and G) and a vacuum pipe system as the cavity X-ray return path. The X-ray diagnostic stations C, D and E are located in the return path and its extension line. Station F is located in the electron beam downstream of crystal C4 of Station B. Station G is located in the LCLS-II experimental station [4].

To keep the optics from intercepting the electron beam in CBXFEL mode, the diamonds and diagnostics are located within 4-dipole electron bunch-delay chicanes that enclose the 7 undulator sections. The electron bunch chicanes divert the electron beam around the inserted optics and must create a beam orbit and deflection in order to maintain the undulator Beam Containment System requirements. The chicane will displace the electron beam by at least 3.914 mm from the undulator axis at the longitudinal location of the diamond insertion [3].

Figure 1 shows a schematic of the CBXFEL cavity with stations A, B, C, D, E and F. Table 1 shows the location of the stations in the SLAC LTU Coordinate system [5].

The return line, shown in Fig. 2, must comply with SLAC's vacuum requirement for LCLS level 2 vacuum. Partial pressure from the sum of all peaks >44 AMU has to be <1x10⁻¹¹ Torr. Maximum single-peak partial pressure for >44 AMU has to be <5x10⁻¹² Torr. The return line comprises 12 ion pumps 75 l/s, 4 pairs of Pirani and Cold Cathode gauges. A new cable tray has been installed in the tunnel to provide signal and power cables to CBXFEL.

Table 1: CBXFEL Coordinates of the Main Components

Station	Z Location [m]	X Location [m]
C1-C2 axis	548.116240	N/A
C3-C4 axis	516.015803	N/A
C1-C4 axis	N/A	-1.25
C2-C3 axis	N/A	-1.90
Station C	546.298572	-1.90
Station D	530.138572	-1.90
Station E	515.249803	-1.90
Station F	517.87823	-1.25

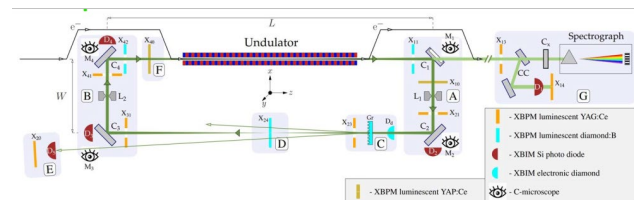


Figure 1: Layout of the x-ray diagnostic design.

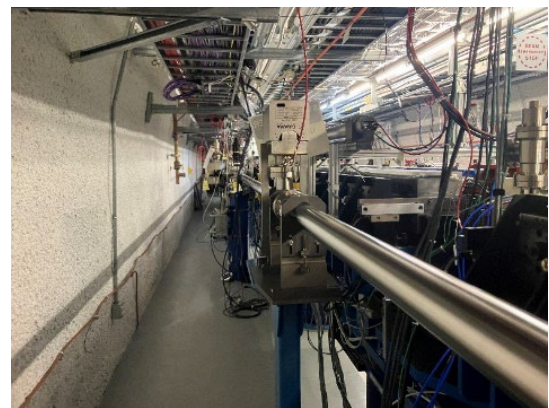


Figure 2: View of the return, with the new cable tray installed on top of CBXFEL.

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X-RAY OPTICS AND DIAGNOSTICS

Station F

Station F is designed to facilitate alignment between the relativistic electron beam entering the undulator line and the X-rays returning from the CBXFEL cavity via the return line. The station features two interchangeable targets, shown in Fig. 3: (1) a boron doped CVD diamond screen for direct imaging of the electron beam, and (2) a YAP:Ce scintillator for detecting returning X-rays in the cavity. Emission from either target is captured using a nanosecond-resolution fast-gated optical/UV camera [6].

Station F will be positioned between the third and fourth dipole magnet of Cell 13. The e-beam will be diverted through the dipole magnets, and the photon beam goes straight, hitting the YAP:Ce crystal, which emits UV light peaking at a 370 nm wavelength. The CVD diamond screen can also be inserted into the e-beam, as well as the photon beam. Through carefully alignment procedures, a minimum 90% beam overlap between the photon and e-beam may be achieved [7].

The Diagnostic shall consist of a BOD card assembly with a vertical actuator, and profile monitoring camera with a horizontal actuator. Vertical actuator has a travel range of 15mm and a resolution of 1.2 μm . Horizontal actuator has a travel range of 25 mm and a resolution of 5 μm .

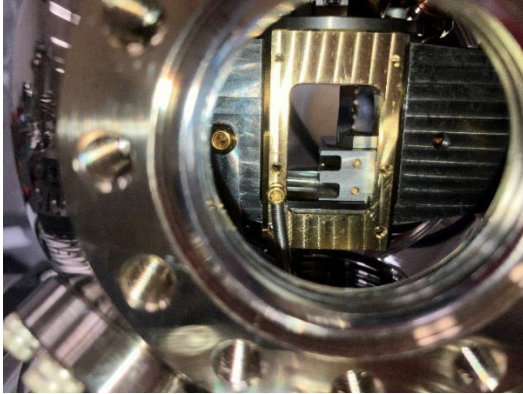


Figure 3: View of the diamond and the scintillator installed in station F paddle.

Station C

Station C is a multipurpose diagnostic station. It consists of three main targets: 1) X23E is an intra-cavity diamond fast sensor, single channel X-ray intensity monitor to be read out by the digitizer, 2) X23G is a diamond grating element which splits the main X-ray beam in three arms (0th, +1st, and -1st orders), with one arm directed to Station E to diagnose alignment, position, and intensity, and 3) X23, a YAG screen as conventional scintillator to evaluate the position of the beams after splitting from the grating and for better angular pointing into C3. The station is installed at a distance of 2-3 m downstream of C2, featuring a field of view limited to 10x10 mm² [8]. Sensors are installed on off-the-shelf nanopositioning XY stages from SmaAct. Travel range is 12 mm, resolution is 50 nm.

Views of X23G, X23E and X23 are shown in Fig. 4.

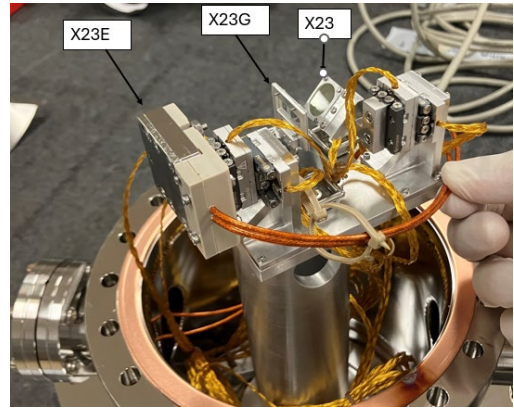


Figure 4: View of Station C with X23G, X23E and X23.

Station D

Station D is a diagnostic with a thin scintillator screen without a central aperture, but with sufficiently low absorption that will be used to enable final precision alignment and beam imaging. Station D grating will also direct $\sim 10\%$ of the x-rays to station E, where detailed information including the x-ray wavefront within the cavity can be measured [1].

Station D has a XBPM X24 mounted on an off-the-shelf preassembled nanopositioning XY stage from SmarAct with a standard reference mark. Travel range for X and Y is 12 mm and resolution is 50 nm

Station E

Station E is a beam diagnostic unit in the CBXFEL setup. It accepts the +1 and -1 order diffracted beam from upstream grating on Station D. These beams can either be directed to the fast diode, or a thin diamond wavefront Talbot grating target, then imaged on a YAG crystal. Zero (0) order beam is cleared from Si crystal and imaged on YAG crystal for position diagnostic. It is also valuable to perform higher energy resolution measurement (~ 10 -20 meV), which is achieved by inserting a monochromator Si (660) or Si (440) [9].

Station E vacuum chamber sits on a granite table. The vacuum chamber support can be adjusted ± 10 mm with 50 μm resolution. A pneumatic gate valve can isolate the vacuum from the return line if needed. Vacuum level requirement is 5×10^{-8} torr or lower. Main motion parameters are listed in Table 2.

Table 2: Basic Motion Control Requirements for Station E

Axis	Travel Range	Resolution
Crystal rotation Bragg angle	-5/95 deg	0.1 μrad
Diode /WFS target X	± 5 mm	1 μrad
Diode /WFS target Y	0/-100mm	1 μrad
Diode /WFS target Z	± 5 mm	5 μrad

Station A and Station B

Station A is installed on sector 21 of the Undulator Hall, and contains two vacuum chambers with crystals C1 and C2. Station B is installed on sector 13 of the Undulator Hall, and contains crystals C3 and C4.

C1, C2, C3 and C4 chamber internals consist of a motion stack mounted on the chamber base flange and an invar alloy reference frame surrounding the motion stack. The motion stack and reference frame provide diamond crystal orientation and position capability in X, Y, Z, roll, and pitch. Motion stack consists of linear flexure stages Z and X, roll and pitch flexure stages, and SmarAct stages (C-Dp, C-Ds, C-Y). Flexure stages have a picomotor for coarse motion and a piezo actuator for fine motion (only for roll and pitch) and built in capacitive sensors for position feedback and closed loop control. Two beam diagnostic systems for C1, C3 and C4, and one for C2, are suspended from the invar reference frame. SmarAct XY nanopositioning stages are used for X and Y positioning the beam diagnostics in the chamber. A gold-coated silicon end target reference mirror is attached to the diamond pitch stage. The end target mirror roll and pitch orientation is measured by two pairs of interferometers and one capacitive sensor for each motion direction. To maintain an even temperature for all the diamond crystals in the cavity system, a heater with a thermostat will be installed on the crystal holder. Main motion parameters are listed in Table 3 [8, 10]. Figure 5 shows a view of the internals of C1.

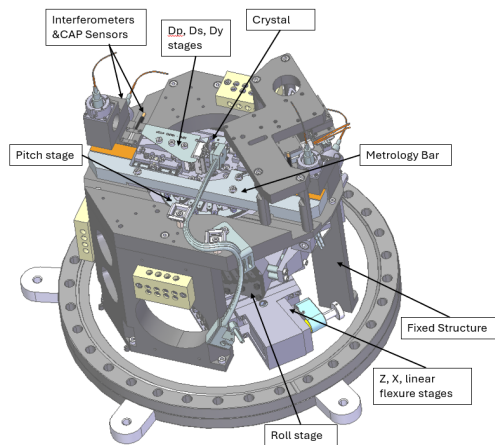


Figure 5: View of C1 with the main components.

Between C1 and C2, and C3 and C4, there will be a nanoposition stage with a x-ray focusing compound refractive lens CRL, installed in the symmetry points of the back scattering units to both control the transverse x-ray beam dynamics and stabilize the cavity.

C1-C2 and C3-C4 are assembled on a base plate. The position of the chambers is defined by the location of alignment pins. A stiffener plate will be installed on top of the girder on sectors 13 and 21, in between the chicane magnets. The stiffener plate will be shimmed and aligned. The base plate is installed on top of the stiffener plate, and a set of pushers allow the alignment of the base plate respect to the stiffener plate.

Station A and B will both have 4 ion pumps, and 4 manual valves to isolate the vacuum of C1, C2, C3 and C4 if needed. A structure has been designed to hold GigE and Andor iStar cameras. The lowest resonant frequency mode of the holding structure shall be over 80 Hz. Figure 6 shows a view of Station A installed in the tunnel.

Table 3: Basic Motion Control Requirements for Station C

Axis	Travel Range	Resolution
Linear Flexure stage Z	1mm	50 nm
Linear Flexure stage X	1mm	50 nm
Coarse pitch	26 mrad	350nrad
Fine stage for pitch	0.1mrad	20nrad
Coarse stage for roll	26mrad	350nrad
Fine stage for roll (fine)	0.1mrad	20nrad
Linear stage for Ds	12 mm	50 nm
Linear stage for Dp	12 mm	50 nm
Linear stage for Dy	12 mm	50 nm

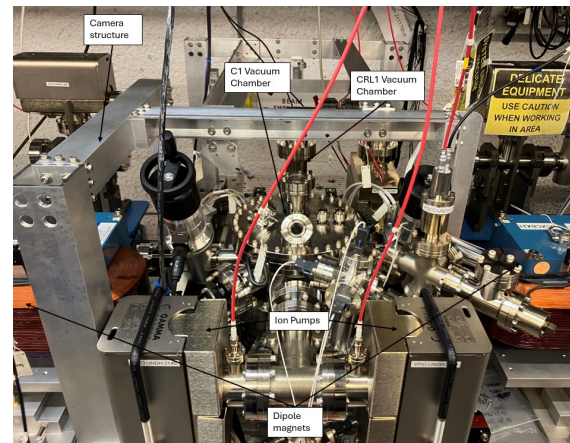


Figure 6: View of Station A installed in the tunnel.

SUMMARY

The CBXFEL experimental is in the final stages of assembly and installation. Installation and final checkout are planned to be checked out by February 2026, followed by startup of the experimental program including measurement of ring-down and amplification.

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